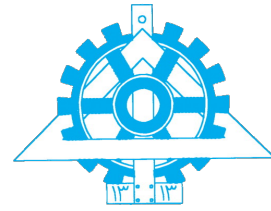




University of Tehran
College of Engineering
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Introduction to Data Science

Assignment 1

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Introduction

This report outlines the findings and visualizations for Data Science Assignment 1. The project encompasses three primary practical tasks: statistical sampling and inference using 2D Gaussian distributions, storytelling through interactive Tableau dashboards for Global Tuberculosis data, and the visual improvement of a financial Power BI dashboard. The programmatic implementation for data processing and mathematical modeling can be found in the accompanying Jupyter Notebook.

Task 1: Sampling and Inference

This section visualizes the comparison between different sampling methods applied to a 2D Gaussian distribution.

2.1 Direct Sampling Visualization

2.2 Langevin Dynamics Sampling

2.3 Comparative Analysis

Task 2: Tableau Dashboard Storytelling

This section presents the interactive dashboards created to analyze the Global Tuberculosis Burden.

3.1 Global TB Overview & Maps

3.2 Key Performance Indicators (KPIs)

3.3 Storytelling and Data Insights

Task 3: Power BI Dashboard Improvement

This section details the enhancements made to the financial Power BI dashboard, focusing on advanced indicators and improved visual hierarchy.

4.1 Original vs. Enhanced Dashboard

4.2 Advanced Financial Indicators

4.3 Design and UX Improvements

Task 4: Data Visualization Principles

5.1 Labeling: Clarity vs. Clutter

5.2 Cognitive Load and Visual Simplicity